7

Linear Maps

Historical Note

The matrix came first and the map came second, which is the reverse of the logical order. Cayley built his algebra in 1858 around arrays of coefficients, and for sixty years afterwards the subject was written that way: a theorem was a statement about entries, and a proof was a calculation. Peano's 1888 book had already defined linear maps between abstract systems, but the change of viewpoint took hold only in the 1920s, in Emmy Noether's Göttingen, where the slogan was that structure lives in the maps and not in the objects, and it reached textbooks through van der Waerden. Paul Halmos wrote *Finite-Dimensional Vector Spaces* in 1942 with the explicit aim of treating finite-dimensional spaces the way Hilbert space is treated, that is, with the operators in front and the matrices behind. We have followed the historical order in this book, building matrices first, because that is easier to learn. This chapter pays the debt.

Chapter 4 defined the matrix product so that matrices would compose like maps, and Chapter 6 ended by turning any n-dimensional space into $\mathbb{R}^n$ through a choice of basis. Putting the two together gives the organising fact of the subject: a linear map between finite-dimensional spaces *is* a matrix, once you say which bases you are using, and the same map becomes a different matrix when you change your mind. What survives the change is what the map really is.

7.1 Linear maps

Definition 7.1 (Linear map)**.** *Let V and W be real vector spaces. A function $T : V \to W$ is* linear *if*

$$T(u+v) = T(u) + T(v) \quad \text{and} \quad T(\lambda v) = \lambda\, T(v)$$

for all $u, v \in V$ and $\lambda \in \mathbb{R}$; equivalently, if $T(\lambda u + \mu v) = \lambda T(u) + \mu T(v)$ always. A linear map from V to itself is called an operator *on V. We write Tv for $T(v)$ when no*

confusion can arise.

Example 7.2 (Linear maps everywhere). *(i)* $T_A : \mathbb{R}^n \to \mathbb{R}^m$, $x \mapsto Ax$, *for a fixed* $m \times n$ *matrix* A. *Linearity is Theorem 4.6. Proposition 7.6 below will show that* every *linear map* $\mathbb{R}^n \to \mathbb{R}^m$ *arises this way.*

(ii) Differentiation $D : P_n \to P_{n-1}$, $p \mapsto p'$. *Linear because* $(\lambda p + \mu q)' = \lambda p' + \mu q'$.

(iii) Integration $I : C[0,1] \to \mathbb{R}$, $f \mapsto \int_0^1 f(t)\,dt$.

(iv) Evaluation $\mathrm{ev}_a : P_n \to \mathbb{R}$, $p \mapsto p(a)$, *for a fixed* a.

(v) Transposition $M_{m\times n}(\mathbb{R}) \to M_{n\times m}(\mathbb{R})$, $A \mapsto A^t$, *by Proposition 4.9.*

(vi) The trace $M_n(\mathbb{R}) \to \mathbb{R}$.

(vii) Rotation of the plane through a fixed angle, and reflection in a fixed line through the origin. Section 7.6 computes the second of these.

Example 7.3 (Maps that are not linear). *The translation* $x \mapsto x + b$ *with* $b \neq \mathbf{0}$ *is not linear, since it sends* $\mathbf{0}$ *to* b*; such maps are called* affine*, and Theorem 3.14 was a statement about them. Neither is* $x \mapsto \|x\|$, *since* $\|-x\| = \|x\| \neq -\|x\|$ *in general, nor* $A \mapsto \det A$ *for* $n \geq 2$, *since* $\det(\lambda A) = \lambda^n \det A$, *nor* $(x_1, x_2) \mapsto (x_1^2, x_2)$. *The word* linear *in "linear regression" refers to linearity in the parameters, not in the inputs, and a model with an intercept term is affine in the inputs; the standard trick of appending a constant* 1 *to every input vector is exactly the device that turns the affine map back into a linear one.*

Proposition 7.4 (Immediate consequences). *If* $T : V \to W$ *is linear then* $T\mathbf{0} = \mathbf{0}$, $T(-v) = -Tv$, *and* $T\left(\sum_i \lambda_i v_i\right) = \sum_i \lambda_i Tv_i$ *for every finite sum.*

Proof. $T\mathbf{0} = T(0 \cdot \mathbf{0}) = 0 \cdot T\mathbf{0} = \mathbf{0}$ by Proposition 5.2(b), and $T(-v) = T((-1)v) = (-1)Tv = -Tv$. The last statement follows by induction on the number of terms. □

The following theorem is small, and it is the engine of the whole chapter.

Theorem 7.5 (A linear map is a free choice on a basis). *Let* $B = (v_1, \ldots, v_n)$ *be a basis of* V *and let* $w_1, \ldots, w_n$ *be* any *vectors of* W*, not assumed distinct or independent or anything else. Then there is exactly one linear map* $T : V \to W$ *with* $Tv_i = w_i$ *for every* i.

Proof. *Existence.* Every $v \in V$ is $\sum_i \lambda_i v_i$ for exactly one list of scalars, by Proposition 6.7,

so the recipe

$$T\Big(\sum_i \lambda_i v_i\Big) := \sum_i \lambda_i w_i$$

defines a function without ambiguity. It is linear: if $u = \sum \alpha_i v_i$ and $v = \sum \beta_i v_i$ then $\lambda u + \mu v = \sum(\lambda\alpha_i + \mu\beta_i)v_i$, whose coordinates are what they must be by uniqueness, so $T(\lambda u + \mu v) = \sum(\lambda\alpha_i + \mu\beta_i)w_i = \lambda Tu + \mu Tv$. Taking all λ_j zero except $\lambda_i = 1$ gives $Tv_i = w_i$.

Uniqueness. If S is linear with $Sv_i = w_i$ then for any $v = \sum \lambda_i v_i$, Proposition 7.4 gives $Sv = \sum \lambda_i w_i = Tv$. □

Key Idea

A linear map costs exactly n arbitrary choices, one per basis vector, and nothing more. Where the basis goes, everything goes. This is why matrices are enough to describe linear maps, and it is worth contrasting with the situation for continuous or smooth maps, where knowing the values at finitely many points tells you almost nothing.

Proposition 7.6 (Every linear map on $\mathbb{R}^n$ is a matrix)**.** *If $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear, then $T = T_A$ for the unique $m \times n$ matrix A whose jth column is Te_j.*

Proof. Let A have columns $Te_1, \dots, Te_n$. Then $Ae_j = Te_j$ by Proposition 4.4, so T and T_A are linear maps agreeing on a basis, hence equal by Theorem 7.5. Uniqueness holds because A's columns are forced to be $Ae_j = Te_j$. □

Proposition 7.7 (The maps themselves form a space)**.** *The set* $\mathrm{Hom}(V, W)$ *of linear maps $V \to W$ is a vector space under $(S + T)(v) = Sv + Tv$ and $(\lambda T)(v) = \lambda Tv$. If* $\dim V = n$ *and* $\dim W = m$ *are finite then* $\dim \mathrm{Hom}(V, W) = mn$.

Proof. The axioms hold pointwise because they hold in W, and sums and scalar multiples of linear maps are linear by a one-line check. For the dimension, fix bases and use Theorem 7.5: a linear map is determined by the n vectors $Tv_1, \dots, Tv_n \in W$, freely chosen, so $\mathrm{Hom}(V, W)$ is in bijection with W^n in a way respecting the operations, and $\dim W^n = mn$. □

7.2 Kernel and image

Definition 7.8 (Kernel and image)**.** *For a linear map $T : V \to W$,*

$$\ker T = \{v \in V : Tv = \mathbf{0}\} \subseteq V, \qquad \operatorname{Im} T = \{Tv : v \in V\} \subseteq W.$$

Proposition 7.9. $\ker T$ *is a subspace of V and* $\operatorname{Im} T$ *is a subspace of W. Moreover T is injective if and only if* $\ker T = \{\mathbf{0}\}$.

Proof. Both contain $\mathbf{0}$ by Proposition 7.4. If $Tu = Tv = \mathbf{0}$ then $T(\lambda u + \mu v) = \mathbf{0}$, and $\lambda Tu + \mu Tv = T(\lambda u + \mu v)$ shows the image is closed under combinations; apply Proposition 5.6.

If T is injective and $Tv = \mathbf{0} = T\mathbf{0}$ then $v = \mathbf{0}$. Conversely if $\ker T = \{\mathbf{0}\}$ and $Tu = Tv$, then $T(u - v) = \mathbf{0}$, so $u - v = \mathbf{0}$. □

For $T = T_A$ these are old friends: $\ker T_A = N(A)$ and $\operatorname{Im} T_A = C(A)$, by Definition 5.11. The point of the abstract versions is that differentiation and integration have them too, and the next theorem applies to those as well.

7.3 Rank and nullity

Theorem 7.10 (Rank–nullity)**.** *Let V be finite-dimensional and $T : V \to W$ linear. Then* $\operatorname{Im} T$ *is finite-dimensional and*

$$\dim \ker T + \dim \operatorname{Im} T = \dim V.$$

Proof. Let $k = \dim \ker T$, which is at most $\dim V = n$ by Theorem 6.14(g). Choose a basis $u_1, \dots, u_k$ of $\ker T$ and extend it, using Theorem 6.14(e), to a basis

$$u_1, \dots, u_k,\ z_1, \dots, z_r \qquad \text{of } V, \qquad k + r = n.$$

We claim $Tz_1, \dots, Tz_r$ is a basis of $\operatorname{Im} T$.

Spanning. Any element of $\operatorname{Im} T$ is Tv for some $v = \sum_i a_i u_i + \sum_j b_j z_j$, and then $Tv = \sum_i a_i \, Tu_i + \sum_j b_j \, Tz_j = \sum_j b_j \, Tz_j$, since each $Tu_i = \mathbf{0}$.

Independence. Suppose $\sum_j b_j \, Tz_j = \mathbf{0}$. By linearity $T\big(\sum_j b_j z_j\big) = \mathbf{0}$, so $\sum_j b_j z_j \in \ker T$ and hence $\sum_j b_j z_j = \sum_i c_i u_i$ for some c_i. Rearranged, this reads $\sum_j b_j z_j - \sum_i c_i u_i = \mathbf{0}$, a relation among the vectors of a basis of V, so every b_j (and every c_i) is zero.

Therefore $\dim \operatorname{Im} T = r$ and $k + r = n$. □

Definition 7.11. $\operatorname{rank} T = \dim \operatorname{Im} T$ *and the* nullity *of* T *is* $\dim \ker T$.

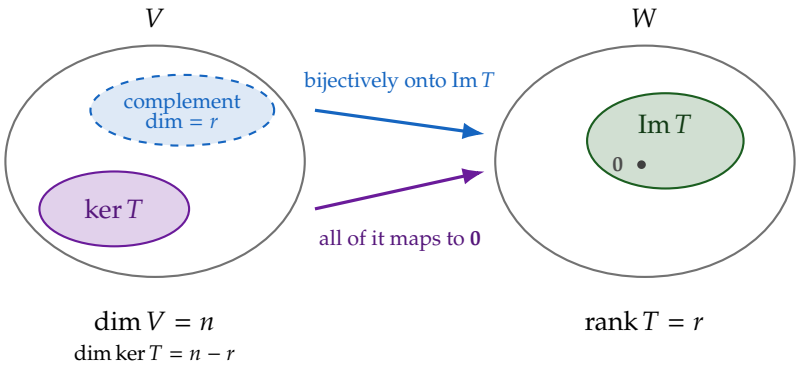


Figure 7.1. Theorem 7.10 is the proof drawn. Split V into the kernel and any complement of it. The kernel is crushed to a single point, and the complement is carried onto the image without loss, one vector for one vector. So the n dimensions of V are accounted for exactly twice over: $n-r$ of them destroyed and r of them preserved. The complement is not canonical, but its dimension is.

Corollary 7.12 (The matrix case, again)**.** *For an* $m \times n$ *matrix* A, $\operatorname{rank} A + \dim N(A) = n$.

Proof. Apply Theorem 7.10 to $T_A : \mathbb{R}^n \to \mathbb{R}^m$, whose kernel is $N(A)$ and whose image is $C(A)$. □

We proved this once already, as Corollary 6.17, by counting pivots. It is worth having both arguments. The pivot count is how you compute the numbers; the present proof is why the identity is true, and it holds for differentiation and integration, where there are no pivots to count.

Corollary 7.13 (Injective, surjective, bijective)**.** *Let* $\dim V = \dim W = n$ *be finite and let* $T : V \to W$ *be linear. Then* T *is injective if and only if it is surjective, and in that case it is bijective.*

Proof. T is injective exactly when $\dim \ker T = 0$, by Proposition 7.9, which by Theorem 7.10 happens exactly when $\operatorname{rank} T = n$. And $\operatorname{Im} T$ is a subspace of W with $\dim W = n$, so by Theorem 6.14(g) it equals W exactly when its dimension is n. □

Caution

This is a theorem about finite dimensions and it fails badly without them. On the space of infinite sequences, the shift $S(x_1, x_2, x_3, \ldots) = (0, x_1, x_2, \ldots)$ is injective and not surjective, while differentiation on $\mathbb{R}[t]$ is surjective and not injective, its kernel being the constants. Corollary 7.13 is the abstract form of Proposition 4.17, and it fails for the same reason.

Example 7.14 (Rank and nullity of differentiation)**.** *Let $D : P_3 \to P_3$ be $p \mapsto p'$. Its kernel is the constants, of dimension 1, and its image is P_2, of dimension 3. Indeed $1 + 3 = 4 = \dim P_3$. Note that D is neither injective nor surjective as a map $P_3 \to P_3$, which is consistent with Corollary 7.13: for an operator on a finite-dimensional space, the two failures always come together.*

7.4 Isomorphism

Definition 7.15. *A linear bijection $T : V \to W$ is an* isomorphism*, and V and W are* isomorphic*, written $V \cong W$, if one exists.*

Proposition 7.16. *If T is an isomorphism then T^{-1} is linear, so T^{-1} is an isomorphism too. Two finite-dimensional spaces are isomorphic if and only if they have the same dimension.*

Proof. For linearity of T^{-1}, let $w_1, w_2 \in W$ and put $v_i = T^{-1}w_i$. Then $T(\lambda v_1 + \mu v_2) = \lambda w_1 + \mu w_2$, so applying T^{-1} gives $T^{-1}(\lambda w_1 + \mu w_2) = \lambda T^{-1}w_1 + \mu T^{-1}w_2$.

If $T : V \to W$ is an isomorphism and B is a basis of V, then $T(B)$ is a basis of W: it spans because T is onto, and it is independent because $\sum \lambda_i T v_i = \mathbf{0}$ gives $T(\sum \lambda_i v_i) = \mathbf{0}$, hence $\sum \lambda_i v_i = \mathbf{0}$ by injectivity, hence all $\lambda_i = 0$. So $\dim W = \dim V$. Conversely, if $\dim V = \dim W = n$, choose bases (v_i) and (w_i) and let T be the unique linear map with $Tv_i = w_i$, from Theorem 7.5; it carries a basis to a basis, so it is onto and, by Corollary 7.13, bijective. □

Key Idea

Up to isomorphism there is exactly one real vector space of each finite dimension, and it is $\mathbb{R}^n$. The coordinate map $v \mapsto [v]_B$ of Proposition 6.20 is the isomorphism, one for each choice of basis. That is the entire content of the phrase "choose coordinates".

7.5 The matrix of a linear map

Definition 7.17 (Matrix of a map). *Let $B = (v_1, \dots, v_n)$ be an ordered basis of V and $C = (w_1, \dots, w_m)$ one of W, and let $T : V \to W$ be linear. The* matrix of T with respect to B and C *is the $m \times n$ matrix*

$$[T]_{C \leftarrow B} \quad \text{whose } j\text{th column is} \quad [Tv_j]_C.$$

When $V = W$ and $B = C$ we write simply $[T]_B$.

Theorem 7.18 (The matrix computes the map). *For every $v \in V$,*

$$[Tv]_C = [T]_{C \leftarrow B}\, [v]_B.$$

Moreover $[T]_{C \leftarrow B}$ is the only matrix with this property.

Proof. Both sides are linear in v: the left by Proposition 6.20 together with the linearity of T, the right by Theorem 4.6. So by Theorem 7.5 it suffices to check them on the basis. For $v = v_j$ we have $[v_j]_B = e_j$, so the right side is $[T]_{C \leftarrow B}\, e_j$, which is column j of the matrix, namely $[Tv_j]_C$, which is the left side. Uniqueness follows because any matrix satisfying the identity has jth column equal to its product with e_j, forcing it to be $[Tv_j]_C$. □

Figure 7.2 is the statement as a square of arrows: across the top and then down is the same as down and then across, and the two vertical arrows, the coordinate maps, are the only arbitrary thing in it.

Theorem 7.19 (Composition is multiplication). *Let $T : V \to W$ and $S : W \to X$ be linear, with ordered bases B, C, D of V, W, X. Then*

$$[S \circ T]_{D \leftarrow B} = [S]_{D \leftarrow C}\, [T]_{C \leftarrow B}.$$

Proof. For every $v \in V$, two applications of Theorem 7.18 give

$$[(S \circ T)v]_D = [S(Tv)]_D = [S]_{D \leftarrow C}\, [Tv]_C = [S]_{D \leftarrow C}\, [T]_{C \leftarrow B}\, [v]_B,$$

and the uniqueness clause of that theorem identifies the product with $[S \circ T]_{D \leftarrow B}$. □

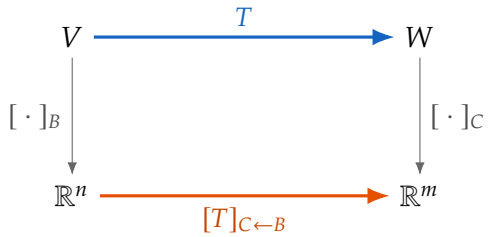


Figure 7.2. Theorem 7.18. The vertical arrows are the coordinate isomorphisms, one for each chosen basis, and they are the only arbitrary thing in the picture. Going across the top and then down is the same as going down and then across, which is what it means to say the matrix represents the map.

Key Idea

This is the theorem Cayley's definition was designed to produce, and we have now come the whole way round. In Chapter 4 the product rule looked arbitrary and was justified by Proposition 4.3, a computation with indices. Here it is a consequence of what a matrix *is*. Any other definition of the product would have made Theorem 7.19 false.

Composing maps multiplies their matrices, and it can also destroy rank. Figure 7.3 shows two maps of rank one whose composite is zero: the first deposits everything into a line that the second sends to the origin, so the loss is visible in the picture rather than in the arithmetic.

Example 7.20 (Differentiation as a matrix)**.** *Take* $D : P_3 \to P_2$ *with the bases* $B = (1, t, t^2, t^3)$ *and* $C = (1, t, t^2)$*. The columns are the coordinate vectors of* $D(1) = 0$*,* $D(t) = 1$*,* $D(t^2) = 2t$ *and* $D(t^3) = 3t^2$*, so*

$$[D]_{C\leftarrow B} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

Check it against $p(t) = 5 - 3t + t^2 + 2t^3$*, whose coordinate vector is* $(5, -3, 1, 2)$*:*

$$[D]_{C\leftarrow B}\,(5, -3, 1, 2)^t = (-3,\ 2,\ 6)^t,$$

which is the coordinate vector of $-3 + 2t + 6t^2$*, and indeed* $p'(t) = -3 + 2t + 6t^2$*. The*

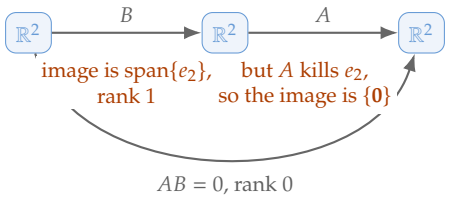


Figure 7.3. Rank can only fall under composition, and it can fall past both factors at once. With $B = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, each map has rank one, and yet $AB = 0$. The reason is visible in the picture rather than in the arithmetic: B deposits everything on the e_2 axis, and that axis is exactly the kernel of A. So the general bound $\operatorname{rank}(AB) \le \min(\operatorname{rank} A, \operatorname{rank} B)$ can be strict on both sides, and equality holds only when the image of B meets the kernel of A trivially. This is also why rank is not multiplicative and why the rank of a product is the hard quantity in Chapter 35, where one wants to build a low-rank matrix as a product and control what it can express.

matrix has one column of zeros, in the position of the constant polynomial, which is precisely $\ker D$*; and it has three pivots, matching* $\operatorname{rank} D = 3$.

7.6 Change of basis and similarity

If the matrix depends on the basis, we had better know how.

Definition 7.21 (Change-of-basis matrix)**.** *Let B and B' be ordered bases of V. The* change-of-basis matrix *from B' to B is $P = [\,\mathrm{id}\,]_{B \leftarrow B'}$, the matrix whose jth column is $[v'_j]_B$, the coordinate vector in B of the jth vector of B'.*

Proposition 7.22. *P is invertible, $P^{-1} = [\,\mathrm{id}\,]_{B' \leftarrow B}$, and $[v]_B = P\,[v]_{B'}$ for every $v \in V$.*

Proof. The identity formula is Theorem 7.18 applied to id. For invertibility, Theorem 7.19 gives

$$[\,\mathrm{id}\,]_{B \leftarrow B'}\,[\,\mathrm{id}\,]_{B' \leftarrow B} = [\,\mathrm{id}\,]_{B \leftarrow B} = \mathbf{1},$$

and the same computation in the other order. □

Theorem 7.23 (Similarity)**.** *Let $T : V \to V$ be an operator, let B and B' be ordered bases*

of V, and let $P = [\,\mathrm{id}\,]_{B \leftarrow B'}$. Then

$$[T]_{B'} = P^{-1}\,[T]_B\,P.$$

Proof. Write $T = \mathrm{id} \circ T \circ \mathrm{id}$ and apply Theorem 7.19 twice:

$$[T]_{B' \leftarrow B'} = [\,\mathrm{id}\,]_{B' \leftarrow B}\,[T]_{B \leftarrow B}\,[\,\mathrm{id}\,]_{B \leftarrow B'} = P^{-1}[T]_B P. \qquad \square$$

Definition 7.24. *Matrices A and A' in $M_n(\mathbb{R})$ are* similar *if $A' = P^{-1}AP$ for some invertible P.*

Key Idea

Similar matrices are the same operator seen in different bases, and nothing else. Every question of the form "how simple can this matrix be made?" is therefore a question about choosing a basis, and the answers occupy Chapters 9 to 11: diagonalisation when a basis of eigenvectors exists, and the Jordan form when it does not.

Example 7.25 (A reflection, in the wrong basis and the right one)**.** *Let T be the reflection of $\mathbb{R}^2$ in the line through the origin spanned by $u = (2,1)$. In the standard basis E its matrix is*

$$[T]_E = \frac{1}{5}\begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix},$$

which shows nothing at a glance. Now take $B = (u, w)$ with $w = (-1,2)$, perpendicular to u. A reflection fixes the mirror and negates the perpendicular direction, so $Tu = u$ and $Tw = -w$, giving

$$[T]_B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The change-of-basis matrix is $P = \left(\begin{smallmatrix} 2 & -1 \\ 1 & 2 \end{smallmatrix}\right)$, whose columns are u and w written in the standard basis, with $P^{-1} = \frac{1}{5}\left(\begin{smallmatrix} 2 & 1 \\ -1 & 2 \end{smallmatrix}\right)$. Verifying Theorem 7.23:

$$[T]_E P = \frac{1}{5}\begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix}\begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 10 & 5 \\ 5 & -10 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix},$$

and then

$$P^{-1}\big([T]_E P\big) = \frac{1}{5}\begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 5 & 0 \\ 0 & -5 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

as promised.

Figure 7.4 draws it. In the standard basis the map is a full matrix of fractions; in the basis adapted to the geometry it is diag(1, −1), one direction fixed and one reversed.

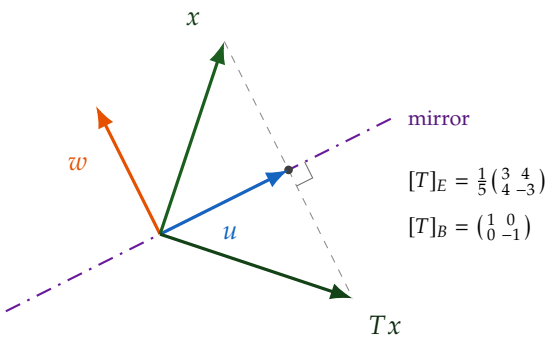


Figure 7.4. Example 7.25. The map is a reflection in the dash-dotted line. Written in the standard basis it is a full matrix of fractions; written in the basis (u, w) adapted to the geometry it is diag(1, −1), which says everything: one direction fixed, one reversed. Here $x = u + w$, so the foot of the perpendicular from x to the mirror is exactly the tip of u, and $Tx = u - w$.

What survives a change of basis

Proposition 7.26 (Similarity invariants)**.** *If* $A' = P^{-1}AP$ *then* $\operatorname{rank} A' = \operatorname{rank} A$ *and* $\operatorname{trace} A' = \operatorname{trace} A$.

Proof. For the rank, note that $A'x = P^{-1}A(Px)$ and that Px runs over all of $\mathbb{R}^n$ as x does, so $C(A') = P^{-1}\big(C(A)\big)$. Since P^{-1} is an isomorphism it carries a basis to a basis, by Proposition 7.16, so the two column spaces have the same dimension. For the trace, first note that $\operatorname{trace}(XY) = \sum_i \sum_j x_{ij}y_{ji} = \operatorname{trace}(YX)$, the two double sums differing only in the order of summation. Hence

$$\operatorname{trace}(P^{-1}AP) = \operatorname{trace}\big((P^{-1}A)P\big) = \operatorname{trace}\big(P(P^{-1}A)\big) = \operatorname{trace} A. \qquad \square$$

Caution

Most things are *not* invariant. The individual entries change, obviously; so does symmetry, since $P^{-1}AP$ need not be symmetric when A is, and so do the lengths of the columns. A quantity is a property of the operator only if you can define it without mentioning a basis, and that is the test to apply. Rank, trace, the determinant of Chapter 8 and the eigenvalues of Chapter 9 all pass it; almost everything you can read off the array at a glance does not.

Insight

The distinction is not pedantry, it is what makes machine learning models comparable. A trained network's weight matrices are written in whatever coordinates the data happened to arrive in, and two networks computing the same function can have completely different weights. Anything you want to say about the function rather than about the parameterisation, effective rank, spectral norm, conditioning, has to be a statement that survives the change. This is also why weight matrices are compared through their singular values, which are invariant under the relevant changes of basis, rather than entry by entry.

Open Problem 7.27 (The Jacobian conjecture)**.** *Let $F : \mathbb{C}^n \to \mathbb{C}^n$ be a polynomial map, so each coordinate of F is a polynomial in $x_1, \dots, x_n$. Its derivative at each point is a linear map, with matrix the Jacobian $JF(x)$. Suppose* $\det JF(x)$ *is a nonzero constant, independent of x.* Must F be invertible, with a polynomial inverse?

What is known. *The converse is easy: if F has a polynomial inverse then the chain rule forces* $\det JF$ *to be a nonzero constant. The forward direction, asked by Keller in 1939, is open for every $n \geq 2$. It is known for $n = 2$ when the degree is at most* 100*, and it is known that the general case reduces to maps of the special form $F(x) = x + H(x)$ with H homogeneous of degree three [53]. That reduction is striking and slightly demoralising: a question about arbitrary polynomial maps in all dimensions is equivalent to a question about cubics, and the cubics are no easier.*

Why this chapter. *The hypothesis is a statement about the derivative, which is a linear map, and the conclusion is a statement about the map itself. It is the sharpest possible test of the principle this chapter is built on, that the linear approximation controls the object. In one variable it is true and trivial: $f'(x)$ a nonzero constant means f is affine. In n variables the same sentence has resisted eighty years.*

Where a reader might start. *Verify the conjecture for $F(x, y) = (x, y + x^2)$, whose Jacobian determinant is* 1*, by writing down the inverse. Then try to build a counterexample: choose H homogeneous of degree two in two variables and impose* $\det J(x + H) = 1$*, and see which coefficients survive. Chapter 8 supplies the determinant identities and Chapter 98 the elimination machinery that current attacks use.*

Exercises

Exercise 7.28. *Which of these maps are linear? Justify each answer.*

(a) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (2x - y,\ x)$.
(b) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (x + 1,\ y)$.
(c) $T : M_n(\mathbb{R}) \to M_n(\mathbb{R})$, $T(A) = A + A^t$.
(d) $T : M_n(\mathbb{R}) \to M_n(\mathbb{R})$, $T(A) = A^2$.
(e) $T : C[0, 1] \to \mathbb{R}$, $T(f) = f(0) - 2f(1)$.

Exercise 7.29. *For* $T : \mathbb{R}^3 \to \mathbb{R}^2$ *given by* $T(x, y, z) = (x + 2y - z,\ 2x + 4y - 2z)$, *find bases for* $\ker T$ *and* $\operatorname{Im} T$ *and verify Theorem 7.10.*

Exercise 7.30. *Let* $T : \mathbb{R}^2 \to \mathbb{R}^2$ *be the unique linear map with* $T(1, 1) = (2, 0)$ *and* $T(1, -1) = (0, 4)$, *which exists by Theorem 7.5. Find* $T(x, y)$ *for a general* (x, y), *and write down* $[T]_E$.

Exercise 7.31. *Let* $S : P_2 \to P_3$ *be* $\big(Sp\big)(t) = \int_0^t p(s)\, ds$. *Write down* $[S]_{C \leftarrow B}$ *for* $B = (1, t, t^2)$ *and* $C = (1, t, t^2, t^3)$, *and compute* $[D]_{B \leftarrow C}[S]_{C \leftarrow B}$, *where* D *is differentiation. Explain the answer, and explain why* $[S]_{C \leftarrow B}[D]_{B \leftarrow C}$ *is not the identity.*

Exercise 7.32. *Show that similarity is an equivalence relation on* $M_n(\mathbb{R})$. *Then show that* $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ *and* $\mathbf{1}_2$ *are not similar, even though they have the same rank, trace and determinant.*

Exercise 7.33. *Let* $T : \mathbb{R}^2 \to \mathbb{R}^2$ *be the orthogonal projection onto the line spanned by* $(2, 1)$, *in the sense of Definition 2.13. Compute* $[T]_E$ *and* $[T]_B$ *for* B *as in Example 7.25, and check that* $T^2 = T$ *in both.*

Exercise 7.34. *On the space of real sequences, let* $S(x_1, x_2, x_3, \ldots) = (0, x_1, x_2, \ldots)$ *be the right shift and* $L(x_1, x_2, x_3, \ldots) = (x_2, x_3, \ldots)$ *the left shift. Show that both are linear, that* $LS = \mathrm{id}$ *and that* $SL \neq \mathrm{id}$, *and say which hypothesis of Corollary 7.13 fails.*

Exercise 7.35. *Verify Proposition 7.7 concretely for $V = \mathbb{R}^2$ and $W = \mathbb{R}^3$ by exhibiting a basis of* $\mathrm{Hom}(V, W)$ *with* 6 *elements.* Hint: Theorem 7.5 and the matrix units of Example 6.8(iii).

Exercise 7.36. *Prove that there is no pair of matrices $A, B \in M_n(\mathbb{R})$ with $AB - BA = \mathbf{1}_n$.* Hint: take the trace of both sides. *Now let D be differentiation on $\mathbb{R}[t]$ and let M be multiplication by t, so $(Mp)(t) = t\,p(t)$. Check that $DM - MD = \mathrm{id}$, and say which step of your argument has no analogue here.*

References and further reading

The map-first presentation of this chapter is the one Halmos made standard, and Axler [37] is its most thoroughgoing modern descendant: he defines everything in terms of operators and postpones matrices as long as possible, which is worth reading alongside our order of presentation to see what each choice costs. The proof of rank and nullity given here, by extending a basis of the kernel, is the usual one, and the picture in Figure 7.1 is the standard way to remember it. Hefferon [307] has a large stock of exercises on writing down the matrix of a map in awkward pairs of bases, a skill that repays practice. For change of basis, for similarity as the organising idea of the subject, and for the view that all of matrix theory is a search for a good basis, Banica [51] is our baseline and takes exactly that line. The reflection of Example 7.25 is a preview of the spectral theorem, where the good basis always exists and is always orthonormal. Treil [610] is worth consulting on the question of why textbooks disagree about which of a matrix and its transpose deserves the name of the matrix of a map.